Jack Schwartz

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Scientific Model Evaluation | Applied Machine Learning | Python

Computational scientist with an M.S. in Computational Science and experience evaluating atmospheric and machine-learning methods with large multidimensional datasets. Designs controlled comparisons, investigates ambiguous results, and builds reproducible Python, Fortran, and HPC workflows for metrics, diagnostics, validation, and clear scientific communication.

TECHNICAL SKILLS

Model Evaluation: comparative benchmarking, baselines, controlled experiments, RMSE, ensemble spread, innovations, analysis increments, distribution diagnostics

Programming & ML: Python, PyTorch, NumPy, pandas, xarray, CNNs, U-Net, Fortran, C++, SQL

Scientific Data: NetCDF, gridded and geospatial data, multidimensional and time-series analysis, validation and QA, scientific visualization

Experimentation & Systems: parameter sweeps, reproducible workflows, configuration management, provenance and logging, technical documentation

Computing & Tools: Linux, Git, GitHub, HPC, Slurm, LLM-assisted research, development, and debugging

RELEVANT EXPERIENCE

Florida State University - Department of Scientific Computing — **Tallahassee, FL**

Graduate Research Assistant — **08/2025 - 08/2026**

- Designed a rigorous head-to-head evaluation of the ensemble score filter (EnSF), LETKF, and a no-assimilation baseline using consistent 80-member ensembles over 30 assimilation cycles.
- Tested linear, fully nonlinear, wind, and partial-observation regimes to separate robust improvement from behavior specific to one observation operator or data-availability setting.
- Evaluated RMSE, ensemble spread, innovations, analysis increments, and state trajectories; EnSF reduced analysis error by up to roughly 20x versus the baseline and led across all variables in the fully nonlinear experiment.
- Investigated inconsistent outputs by tracing intermediate state updates, observation transformations, configurations, and ensemble trajectories before drawing conclusions.
- Automated model-selection sweeps with up to 25 concurrent Slurm jobs, isolated configurations, version-controlled code, and per-run logs to support reproducibility and provenance.
- Documented shared research-computing workflows, mentored a graduate researcher, and presented methods, results, limitations, and caveats to internal and visiting scientists.

Undergraduate Research Assistant — **09/2024 - 05/2025**

- Built Python and Fortran workflows to evaluate free-running 1,000-member global atmospheric ensembles across seasons, variables, pressure levels, and spatial scales.
- Applied spread, skewness, kurtosis, and Gaussianity diagnostics across a 96 x 48 latitude-longitude grid and validated analysis used in a peer-reviewed publication.

PUBLICATION

Atmospheric Science Letters — **05/2026**

Wiley (Royal Meteorological Society)

Co-author, "The Multiscale Non-Gaussian Statistics of Free-Running 1000-Member General Circulation Model Ensembles." doi:10.1002/asl2.70048

EDUCATION

M.S. in Computational Science — **08/2025 - 08/2026**

Florida State University - Department of Scientific Computing — **Tallahassee, FL**

GPA: 4.0

B.S. in Computational Science — **08/2021 - 05/2025**

Florida State University - Department of Scientific Computing — **Tallahassee, FL**

GPA: 3.74 (Magna Cum Laude)

Dean's List: Fall 2021; Spring/Fall 2023; Spring/Fall 2024; Spring 2025

SELECTED TECHNICAL PROJECTS

Medical Image Reconstruction
Python, Machine Learning, U-Net, Model Evaluation

- Implemented U-Net architectures for reconstruction of complex k-space MRI data from the fastMRI breast dataset.
- Prepared training data and evaluated reconstruction quality through a reproducible Python workflow, extending model-evaluation experience beyond atmospheric science.

Automatic Left-Atrium Segmentation from 3D MRI
Python, PyTorch, 3D U-Net, NIfTI, Dice Loss, Interactive Dashboard

- Built a PyTorch pipeline to segment the left atrium from 3D NIfTI MRI volumes using a 3D U-Net, custom Dataset, and training/validation split.
- Implemented spatial preprocessing and dataset normalization, then visually verified that MRI volumes and segmentation labels remained aligned.
- Tracked training and validation loss, retained the best checkpoint, and built an interactive dashboard to review learning curves, architecture, and predictions.
- Evaluated results with Dice-based loss and input, ground-truth, and prediction overlays while accounting for overfitting risk from a 20-case training dataset.

Ensemble Data Assimilation Benchmark
Python, Fortran, SPEEDY, AMLCS-DA, NetCDF, Slurm, Git

- Implemented the first ensemble score-filter application in the SPEEDY atmospheric general circulation model and selected LETKF and no-assimilation baselines for comparison.
- Designed controlled linear, fully nonlinear, wind, and partial-observation experiments using consistent 80-member ensembles and 30-cycle evaluations.
- Compared results across variables and regimes rather than relying on a single headline metric; documented where improvements held and where interpretation required caution.
- Diagnosed failures and inconsistent outputs by tracing state updates, observation transformations, intermediate metrics, and ensemble trajectories.

Scientific Model Evaluation and Diagnostics Toolkit
Python, Fortran, xarray, NetCDF, NumPy, pandas, SPEEDY, AMLCS-DA, Slurm, Git

- Built reusable workflows that transformed per-cycle atmospheric model output into standardized RMSE, ensemble-spread, innovation, and analysis-increment datasets.
- Automated consistent comparisons across methods, variables, model levels, spatial grids, and analysis cycles using shared configuration and processing logic.
- Added schema and value checks, error reporting, heatmaps, time series, and trajectory plots to expose data-quality issues and unexpected model behavior quickly.

CERTIFICATION

4D AI Fluency Certificate **2026**
Training in clear task specification, collaboration with generative-AI tools, output evaluation, and responsible use.

ADDITIONAL EXPERIENCE

Habitat for Humanity ReStore **Stuart, FL**
Store Associate **06/2025 - 08/2025**
Supported nonprofit retail operations, customer service, merchandise organization, and an orderly sales floor while collaborating with staff and volunteers.

Spritz City Bistro **Stuart, FL**
Dining Room Busser **05/2022 - 08/2022**
Coordinated with servers and hosts to maintain efficient front-of-house operations and a positive guest experience in a fast-paced environment.

Eight to Eighty Eyewear **Bellmore, NY**
Warehouse Worker **08/2020 - 08/2021**
Supported filing, quality assurance, shipping, and receiving in a collaborative warehouse environment.